

# Molecular Dynamics

September 9, 2026

## 1 Introduction

Molecular Dynamics can be used to study biomolecules, especially their behaviour and usefulness in a lot of field such biophysics and medicine. Lately a hot topic is the study of biopolymers and their ability to translocate through pores in biological membranes. Processes like the translocation of mRNA, DNA injection by viruses and drug delivery [1] are just some of the processes that involve this phenomenon. In this small study we tackle the understanding of polymers' translocation through the evaluation of the free energy profiles associated. The simple model implemented will be composed of an homopolymer placed through a membrane's pore. We will follow the paper '*Free Energy Evaluation in Polymer Translocation via Jarzynski equality*' [1] and we will implement similar molecular dynamics calculations using LAMMPS [2, 3]. To start, we study the implementation of a polymer in LAMMPS following useful documentation such '*Protocols for Translocation Processes of Flexible Polymers through a Pore using LAMMPS*' [4]. Then we evolve the system as if it was immersed in a thermal bath. Lastly, to evaluate the free energy profiles we implement a method that exploits the so called Jarzynski Equality [5] that requires switching off the thermal and dissipative forces and turn on an external time-dependent driving force applied to the polymer. It is possible to consult all the codes used at the GitHub page [6].

## 2 Methods

To study this phenomenon, in this project the workflow will be the following. First, implement the necessary structure. Second, optimize the cell size and duration of the equilibration phase. After this, we can start the system with a minimization, then proceed with the equilibration with a thermal bath and finally, after switching off the thermal bath and turning on the time-dependence of the external drive, start the actual production from which we will derive the free energy profiles.

### 2.1 Implementation

First of all, we clarify that we will use the Lennard Jones units (lj units from now on). Remember that these units are specified in relation to the parameters  $\sigma$  and  $\epsilon$  that identify the typical length and energy associated to the system.

The aim is to shape an homopolymer and a mono-atomic linear membrane in a 2D cell.

This membrane will cover the entire height of the cell, from its bottom to its top, in such a way that its beads have the same x coordinate ( $x = 0$ ) and are distanced, in the y axis, by a constant value of  $1.1224\sigma$  lj units. Only at position (0,0) a membrane's bead is removed to create the pore.

The polymer will be made out of  $N$  monomers. Initially, this structure will be placed along a 1D line perpendicular to the membrane. Therefore the beads will all have the same  $y$  coordinate ( $y = 0$ ) and they will be enumerated starting from the left to the right, from 1 to  $N$ . At the beginning, one of these beads, that from now on will be called the 'middle monomer', will occupy  $(0, 0)$  and its associated number will be  $n$ . Therefore, the side of the cell on the left of the membrane, also called 'cis' side, will contain  $n - 1$  monomers, whereas the right side, also called 'trans' side, will contain  $N - n + 1$  beads. Remember that the middle monomer is placed in the exact transition point between the cis and trans side. We made the decision to count it as the first monomer in the trans side. This is visible in Figure 1. From now on, the number of monomers in the trans side will be referred to as  $m$  and this number will be used as the control parameter for the free energy profiles.

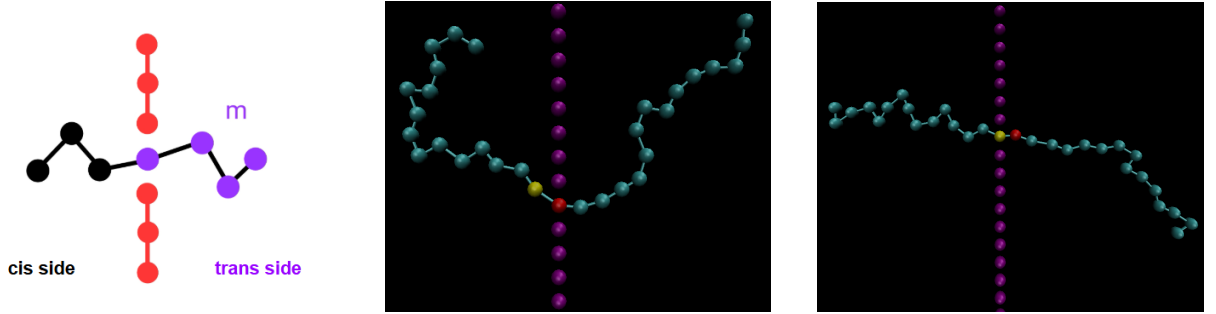


Figure 1: Schematic description of the setup and view of the initial and final step of the production phase from VMD [7]

Let us use now define the potentials and forces used.

- Cutoff Shifted Lennard Jones potential

$$U_{lj}(r) = \begin{cases} 4\epsilon \left[ \left( \frac{\sigma}{r} \right)^{12} - \left( \frac{\sigma}{r} \right)^6 \right] + \epsilon, & r \leq 2^{1/6}\sigma \\ 0, & r > 2^{1/6}\sigma \end{cases} \quad (1)$$

- Finite-Extension Nonlinear Elastic (FENE) potential

$$U_{FENE} = -\frac{1}{2}kR_0^2 \ln \left[ 1 - \left( \frac{r}{R_0} \right)^2 \right] \quad (2)$$

$$\text{where } k = 7\epsilon/\sigma^2, \quad R_0 = 2\sigma$$

- Thermal bath, described by the forces

$$\vec{F}_i^{s,d}(t) = -\xi \frac{d\vec{r}_i}{dt} + \vec{F}_i(t) \quad (3)$$

$$\text{where } \langle \vec{F}_i(t) \rangle = 0 \quad ,$$

$$\left\langle [\hat{n} \cdot \vec{F}_i(t)] [\hat{n}' \cdot \vec{F}_j(t')] \right\rangle = 2\hat{n} \cdot \hat{n}' k_B T \xi \delta_{ij} \delta(t - t') \quad ,$$

$$k_B T = 1.2\epsilon \quad , \quad \xi = 0.7$$

- External time-dependent potential

$$U(r, t) = A e^{-br^2} (\sin[cx - \lambda(t)])^2 \quad (4)$$

$$\text{where } \lambda(t) = \begin{cases} \pi \nu t, & \text{in the Production phase} \\ 0, & \text{in the Equilibration phase} \end{cases} ,$$

$$dt = 3 \cdot 10^{-3} \sqrt{m\sigma^2/\epsilon} \quad , \quad A = 40\epsilon \quad , \quad b = 5/\sigma^2 \quad ,$$

$$c = 3.14/\sigma \quad , \quad \nu = 2.5 \cdot 10^{-3} \sqrt{\epsilon/m\sigma^2}$$

When the bath will be present, it will keep the temperature such that  $k_B T = 1.2\epsilon$  and it will describe stochastic and dissipative forces acting on every bead. Whereas, the external force, that will always be present, will be exerted only on the polymer. Notice that, however, the shape of the external potential is peaked around the point (0,0) and exponentially decreased to zero away from it in all directions. A 1D plot is visible in Figure 4. Such shape implies that only the polymer's beads close to the origin will actually feel the external force. Also, the time dependence of this potential is essential to derive the free energy profile thanks to the Jarzynski equality. In the first part of the dynamics, called Equilibration, it will be fixed in time but still present. In this phase it's duty is to keep the middle monomer in position. In the 'active part', where the potential is actually time-dependent, this external force drives the polymer to a new configuration and thanks to the study of the work done on the system during this process we can evaluate the free energy difference. Finally notice that the parameters described up to now are identical to the ones in the article '*Free Energy Evaluation in Polymer Translocation via Jarzynski equality*' [1]. The parameters of the time-dependent potential,  $A$ ,  $b$  and  $c$ , were unfortunately not specified in the article. Therefore  $b$  and  $c$  have been set such that the minima of the potential at rest are at a distance of the order of the unity in units of  $lj$ . Whereas the amplitude  $A$  is set at the minimum value that allows the potential to actually act as an optic trap on the middle monomer. The one dimensional plot of the external potential is visible in Figure 4, in the Appendix.

The polymer's beads will interact through both LJ and FENE potentials and are initially distanced by  $1.1224\sigma$  units  $lj$  along the  $x$  axis. The membrane's beads will interact only through LJ potential and, for simplicity, they will be kept in position by always imposing that their velocities and forces are zero, after LAMMPS [2, 3] calculated the pairwise interactions between all beads.

## 2.2 Optimization

The next step is to optimize the necessary parameters. First we look at the cell size. In order to optimize it we will embed the system in a square cell of typical length  $2L$ . The system will be positioned around  $(0,0)$  and the cell, in all 2 directions, starts at  $-L$  and ends at  $L$ . Then,  $L$  will be made vary from  $20\sigma$  to  $40\sigma$  with step  $10\sigma$ . We will then run 30 independent equilibrations for each of these values and look at the temperature, energy, x position of the middle monomer, end-to-end distance and radius of gyration of the system, or polymer, over 5 million steps. We will analyse their mean value and whether this changes substantially while lowering the cell size. The lowest value of the characteristic length for which the study shows accordance with the  $L = 40\sigma$  simulation will be taken as the optimized value.

Of course parameters such temperature and energy will start from a random point and equilibrate around their final mean value after some simulation time. Therefore this study is deeply intertwined to the analysis of the minimum simulation time to wait in order to be able to state that the system is actually at equilibrium.

## 2.3 Workflow

As said before, the workflow is made of 3 main phases: Minimization, Equilibration and Production. The first step only ensures that no bead is super positioned and no interaction energy between two beads is too big.

In the next step, we start the equilibration, in which, for the first time, we make our beads move following the applied potentials and thermal bath. This description is achieved with a 'nve' integrator and a 'Langevin' thermal bath. This is an essential step: the application of the Jarzynski Equality assumes that the initial state, that will evolve through the external drive, is drawn from an equilibrium canonical ensemble. Thanks to the bath, in fact, we fix the temperature of the system, apart from obvious numerical fluctuations. Notice that in this stage the external potential is fixed in time and acts like an optical trap, keeping the middle monomer in position.

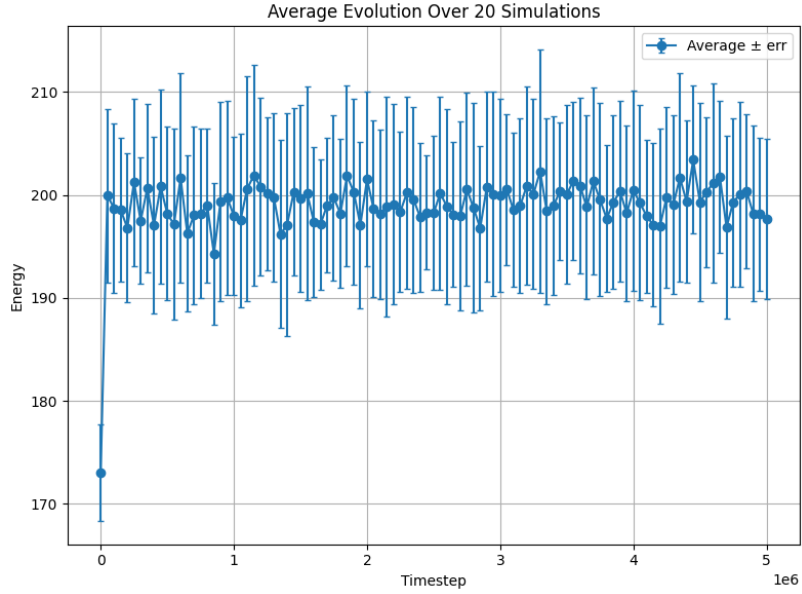
Finally, as described in Jarzynski's paper, '*Nonequilibrium Equality for Free Energy Differences*' [5], for simplicity we can study a deterministic evolution. In this case, described by the absence, in the Production phase, of stochastic and dissipative forces, i.e. the absence of the thermal bath, we can obtain the work done on the system directly from the energy variation. Therefore, between the end of the Equilibration and the start of the Production we switch off the thermal bath and leave only the others conservative potentials. Notice, also, that in this moment the external potential actually becomes time-dependent. We will collect the Energy value from the last Equilibration step, or equivalently the  $0^{th}$  step of the Production, and call it  $E_i$ . The Production lasts 133333 steps, equal to  $400 \sqrt{m\sigma^2/\epsilon}$  (lj time units), since the time step has been set to  $3 \cdot 10^{-3} \sqrt{m\sigma^2/\epsilon}$ . In this time the External potential moves towards the trans side, bringing with it the middle monomer and stops when it recovers for the first time its original shape. In this way the monomer on the immediate left of the central one (the last in the cis side) ends up as the new middle monomer. At the end we collect the final energy  $E_f$  and with it we can calculate the free energy difference thanks to Equation 5.

$$\langle e^{-\beta(E_f - E_i)} \rangle = \langle e^{-\beta W} \rangle = e^{-\beta \Delta F} \quad (5)$$

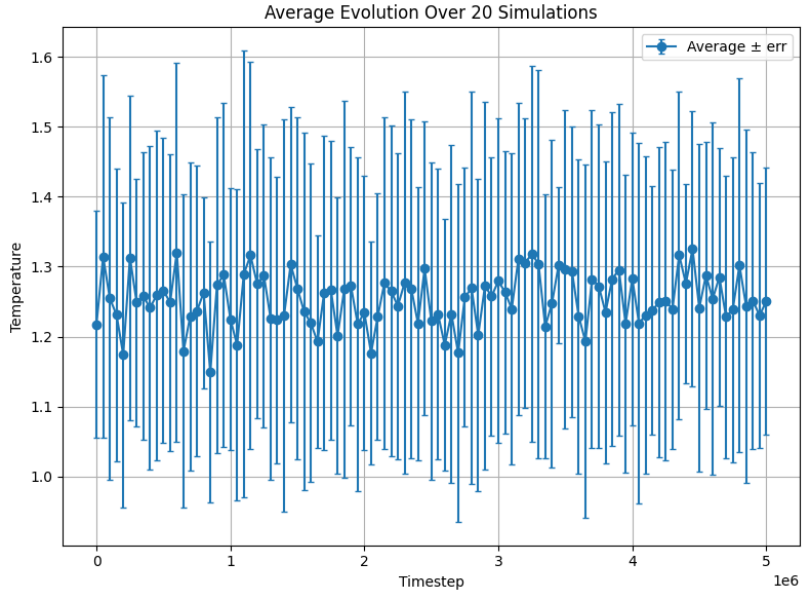
## 3 Results

### 3.1 Optimization

First of all, we study temperature and energy throughout the Equilibration for  $L = 40\sigma$ , to analyse the time it takes the system to reach stability. In Figure 2 we can see the data coming from this Equilibration phase, collected every 50000 steps. From this images it is clear that after 1 million steps the system is already surely stable. For this reason the next averages of the various characteristics of the system studied to optimize the cell size will be estimated from the one million step to the last (ergo over 4 million steps). Moreover, this allows us to fix the Equilibration time of the final study to 1 million steps.



(a) Energy (units of lj) variation along the Equilibration with  $L = 40\sigma$



(b) Temperature (expressed with the value of  $k_B T$ ) variation along the Equilibration with  $L = 40\sigma$

Figure 2

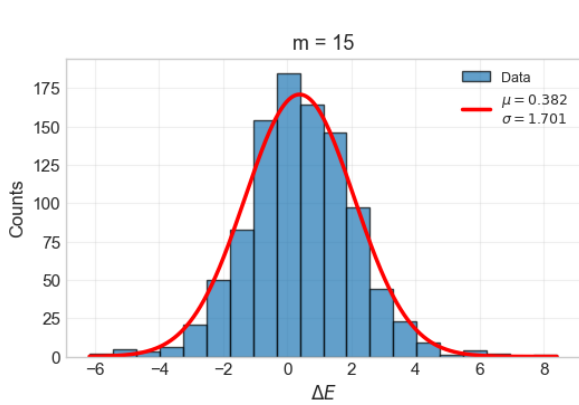
The data coming from the study of the effect of the cell size on temperature, energy,  $x$  position of the middle monomer, end-to-end distance and radius of gyration are summarized in Table 1. It is visible that the cell size does not have a huge impact on these variables and therefore it has been decided that the value  $L = 30\sigma$  was more than enough to let the system evolve undisturbed.

| $L$ | $T$  | $\sigma_T$ | $E$ | $\sigma_E$ | $x$   | $\sigma_x$ | $E2E$ | $\sigma_{E2E}$ | $R_g$ | $\sigma_{R_g}$ |
|-----|------|------------|-----|------------|-------|------------|-------|----------------|-------|----------------|
| 40  | 1.25 | 0.04       | 199 | 1          | -0.01 | 0.02       | 15.7  | 0.6            | 5.7   | 0.1            |
| 30  | 1.25 | 0.04       | 199 | 1          | -0.01 | 0.02       | 15.7  | 0.6            | 5.7   | 0.1            |
| 20  | 1.25 | 0.04       | 199 | 2          | 0.09  | 0.01       | 15.5  | 0.7            | 5.7   | 0.1            |

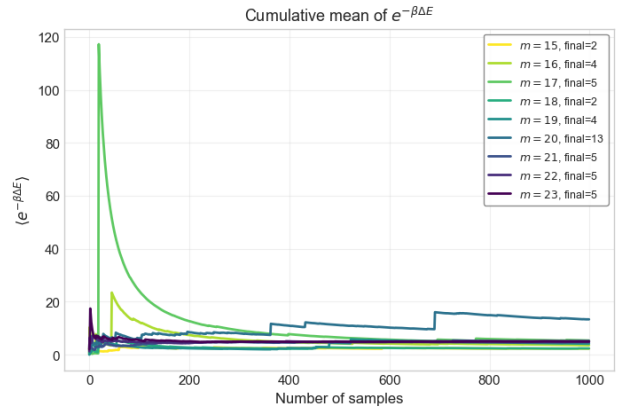
Table 1: Optimization results for different system sizes  $L$ . ' $T$ ' stands for Temperature, described by the value of  $k_B T$ , in units of  $\epsilon$ . ' $E$ ' stands for Energy, in units of  $\epsilon$ . ' $x$ ' stands for the  $x$  position of the middle monomer, in units of  $\sigma$ . ' $E2E$ ' stands for the end to end distance in units of  $\sigma$ . ' $R_g$ ' stands for Radius of Gyration, in units of  $\sigma$ .

### 3.2 Free Energy Profiles

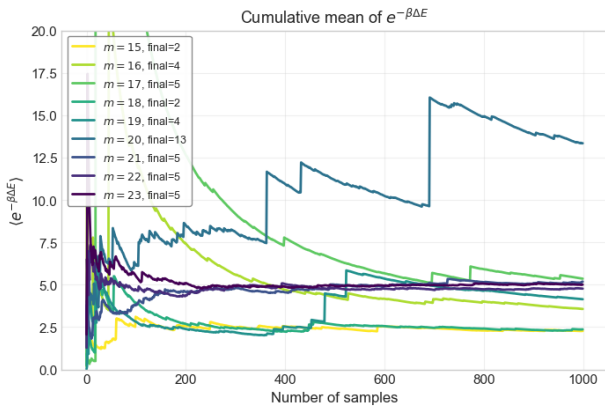
We have analysed a polymer with 30 monomers thanks to a 1000 simulations per value of  $m$ .  $m$  takes unitary values from 15 to 23. The results are show in Table 2 and Figure 3.



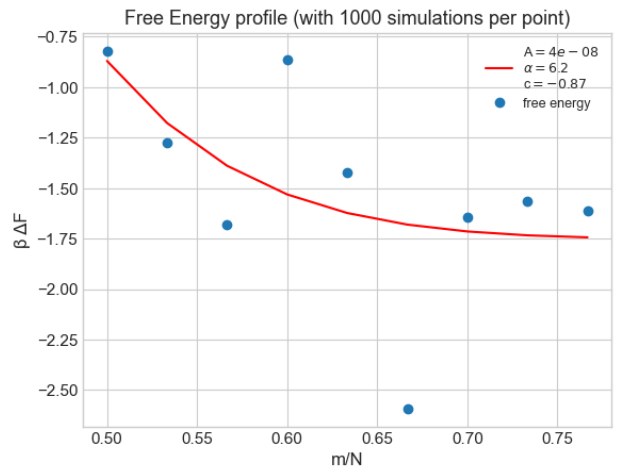
(a) Distribution of Delta Energies for  $m=15$



(b) Cumulative Mean over the 1000 simulations



(c) Closer look at the Cumulative Mean over the 1000 simulations



(d) Free Energy profile, fitted with Eq. 6

Figure 3: Results for the 2D 30 monomers system

| $m$ | $\langle e^{-\beta\Delta E} \rangle$ | $\beta\Delta F$ |
|-----|--------------------------------------|-----------------|
| 15  | 2.27151485                           | -0.82044694     |
| 16  | 3.57000614                           | -1.27256732     |
| 17  | 5.35619723                           | -1.67825425     |
| 18  | 2.36612580                           | -0.86125393     |
| 19  | 4.14396633                           | -1.42165338     |
| 20  | 13.34753761                          | -2.59133192     |
| 21  | 5.18238417                           | -1.64526522     |
| 22  | 4.77007779                           | -1.56236261     |
| 23  | 5.00569799                           | -1.61057686     |

Table 2: Results for the 2D 30 monomers system

Following the article [1] and to make the free energy profile tendency clear, in the last plot (Figure 3d) we fitted the data using Equation 6.

$$\beta\Delta F = A \left[ (N - m)^\alpha - \left( \frac{N}{2} \right)^\alpha \right] + c \quad (6)$$

The free energy profile is somewhat decreasing, which is expected, but the data seem to be scattered, as if a thousand simulations wasn't enough for the system to reach a sort of stability. This is confirmed by Figure 3c where it is visible that the mean is not yet stable to the appearance of largely negative delta energies. Notice that this study is based on an out of equilibrium driving, meaning we abandon the canonical ensemble in the production phase. This implies that the distribution of the delta energies should not be Gaussian: as shown in Figure 3a, there are values in the tail of the fitted Gaussian that are rather unlikely. This means that we expect lots of delta energies in the positive tail as well as in the negative one. The latter is obviously more important, seen the negative exponential of equation 5. Therefore we not only want those negative values but our entire study is based on them. So, the fact that our mean still appears to be sensible to these new values in the negative tail suggests that we need even more simulations to actually reach the real free energy profile. Moreover in this study we used a 2D implementation of the 3D model studied in the article [1]. In such a system, the role of entropy diminishes a lot, compared to a higher dimensional one. This characteristic should make the number of required simulation decrease in our model. All these informations rise questions upon why the team of Mondaini and Moriconi [1] were able to find such perfectly decreasing free energy profiles in only one hundred simulations per point. A possibility to be studied is the role of the length of the polymer, that in our system was 40 % shorter.

## 4 Appendix

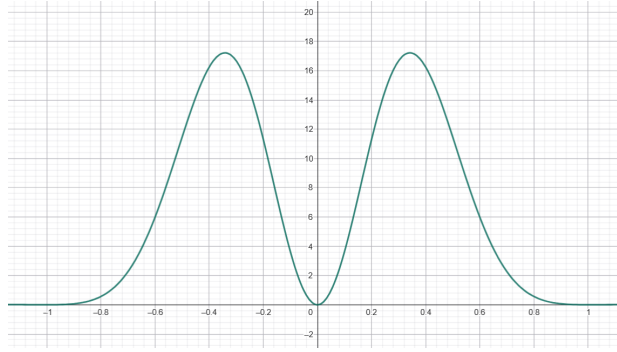


Figure 4: *External Potential shape*

## References

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